MERA

Mirror Entity Recursive Agent

Self-Healing AI Agent Powered by SigNoz Observability

Track 03 - Build Your Own

Agents of SigNoz Hackathon 2026 | SigNoz x WeMakeDevs

Team Enthusiast

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1. Kya? (What is MERA?)

MERA ek self-healing AI agent system hai jo apne aap ko observe karta hai, mistakes detect karta hai, aur khud ko fix karta hai - without human intervention.

Core Concept

Recursive observability loop: Main Agent works with OTel, Mirror Agent reads traces via SigNoz MCP, detects anomalies, generates fixes. Everything visualized on SigNoz dashboards with alerts.

Components

- Main Agent (PR Reviewer): Llama 3.2 (local) code review, OTel-instrumented
- Mirror Agent (Observer/Healer): MCP queries, anomaly detection, fix generation
- SigNoz Platform: All 6 features - Traces, Metrics, Logs, Dashboards, Alerts, MCP
- Self-Healing Loop: Work -> Watch -> Detect -> Fix -> Update

2. Kyu? (Why MERA?)

The Problem

AI agents are black boxes. LLM hallucinations, latency spikes, cost explosions - all invisible. Traditional tools show what broke but not why or how to fix.

Pain Points

- AI agents chain LLM calls, tools, vector DBs autonomously - all opaque
- Latency, cost, hallucination issues invisible until users complain
- Fragmented observability (different tools for logs/metrics/traces)
- No automated self-healing - manual debugging every time

Why SigNoz?

OpenTelemetry-native (agents understand OTel), one platform for all signals, MCP server for programmatic access, open source (no lock-in), consistent schema across traces/metrics/logs.

Mission

"If you cannot observe your AI agents, you do not own them." MERA proves AI systems can be self-aware, self-debugging, and self-healing with proper observability.

3. Kese? (How is it built?)

Three layers: Agent Layer (Main + Mirror), Observability Layer (SigNoz + OTel Collector), Infrastructure Layer (Docker + Foundry).

Tech Stack

Language: Python 3.11+

AI/LLM: Ollama + Llama 3.2 3B (local, free, no API key)

Observability: OpenTelemetry SDK + OTLP HTTP Exporter

SigNoz: Traces + Metrics + Logs + Dashboards + Alerts + MCP

Deployment: Docker + SigNoz Foundry (casting.yaml)

Main Agent (agent.py)

- Ollama OpenAI-compatible API se PR review agent with OTel spans
- Custom attributes: code.language, review.confidence_score, review.issues_count, llm.latency_ms
- Anomaly score from recent history (avg confidence, issue count)
- Robust JSON parsing with regex fallback for LLM output

Mirror Agent (mirror.py)

- SigNoz MCP se traces/alerts fetch via JSON-RPC
- 3 anomaly rules: high latency (>5s), low confidence (<0.5), zero-issues-suspicious
- LLM-based fix suggestion generation per anomaly
- Auto-creates SigNoz dashboards + alert rules via MCP API
- Self-OTel-instrumented (meta-recursive)

Orchestrator (run.py)

- Main agent continuous loop (3 sample codes cycling)
- Mirror agent 3 self-healing cycles with 8s intervals
- Threading for concurrent execution

4. Kab? (Timeline)

Hackathon: July 20 - July 26, 2026

Registration: https://forms.gle/uxaLXAXmtKwz8uYh9

Pre-Hackathon

- Ollama install, SigNoz Foundry setup, OTel learn

Day 1 (Jul 20)

- Main Agent code + OTel instrumentation + test

Day 2 (Jul 21)

- Mirror Agent + MCP integration + anomaly detection

Day 3 (Jul 22)

- Self-healing logic + fix generation + orchestration

Day 4 (Jul 23)

- Dashboards + Alerts + Docker + Foundry deploy

Day 5 (Jul 24)

- Tests + README + architecture diagram + blog

Day 6 (Jul 25)

- Demo video (2 min) + submit + social media

Day 7 (Jul 26)

- Buffer / Judge Q&A prep

5. Tools Used & Free Status

Tool	License	Free?	Notes
Ollama	MIT	Fully Free	Local LLM, no API key needed
Llama 3.2 3B	Meta OSS	Fully Free	2GB model, runs locally
SigNoz (Self-Host)	MIT	Fully Free	Local Docker deploy
OpenTelemetry SDK	Apache 2.0	Fully Free	CNCF standard
Python 3.11+	PSF	Fully Free	Language runtime
Docker	Freemium	Free personal	Desktop free tier
Foundry CLI	Open Source	Fully Free	npm package
GitHub	Freemium	Free unlimited	Public repos
AWS Credits	Sponsored	$100/participant	Builder Center signup
pytest	MIT	Fully Free	Testing
fpdf2 (Report)	LGPL	Fully Free	PDF generation

Cost estimate: $0. Zero. Ollama local hai, SigNoz self-host free hai, AWS $100 free credits mil rahe hain. Ek bhi paisa nahi lagana padega.

6. Judging Score Card

Criteria	Wt	Score	Explanation	
Potential Impact	20%	18/20	Self-debugging AI = trillion-dollar problem. -2 for use-case scope.	
			Recursive self-watching agent never done in hackathon. Maximum.	Creativity
			OTel+MCP+LLM clean. -5 for edge cases (agent corruption).	Technical E
			All 6 features: Traces+Metrics+Logs+Dashboards+Alerts+MCP. Max.	Best Use o
			Auto-dashboards help. -6 because concept complex for non-tech judges.	User Exper
			Strong docs+diagram+tests+script. -3 for production polish.	Presentatio

Final Estimated Score: 84 / 100

Top 3 almost certain | Win probability: 70% | iPhone 17 Pro Max dono ko

7. What Makes Team Enthusiast Win

Unfair Advantages

- 6/6 SigNoz features used - most teams use 2-3 max
- Recursive self-watching concept - extremely novel
- Full OTel instrumentation - every step traced
- Maximum MCP use - agents talk to observability data programmatically
- Foundry deploy - casting.yaml makes it reproducible (rules requirement)
- Team of 2 = focused execution, no coordination overhead
- Zero cost - Ollama local LLM, no API key needed
- Auto-generated dashboards + alerts impress judges

Prize Breakdown per Member

- Apple iPhone 17 Pro Max (or cash equivalent) - Track 3 winner
- AWS credits: $5K (1st) / $3K (2nd) / $2K (3rd)
- Job interviews at SigNoz
- LEGO Ferrari SF-24 (best blog side track)
- Exclusive swag (social media top 10)

Strong Concept + Deep SigNoz + Clean Code + Great Demo = iPhone 17 Pro Max

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Team Enthusiast | Dr. Kiran & Pallavi Patel Global University, Vadodara